

CSCI 8605 3D Drawing in Extended Reality

Final Report

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1. Weekly Drawing Retrospective

I did not think of myself as someone who would actually make nice drawings. I never took any art or drawing class, and I was not sure if I would be able to adapt and improve my drawing skills. In the first weeks, it was genuinely difficult, learning to draw in VR while also just learning to draw at all was a lot to take on at once. But with each week and each assignment, I started to notice real differences in what I was producing.

The first task, I remember struggling just to find a starting point. Eventually I tried something with ASCII-style shapes and drew a simple cat to see how it would look and I actually liked the result. (Figure 1)

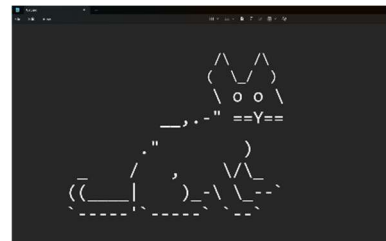


Figure 1. Drawing attempt using ASCII-style shapes

My first actual VR drawing was frustrating in a different way. I spent far more time than I expected on what seemed like a simple prompt, and nothing I made felt right. Then I stopped, cleared everything, and tried something much simpler, a rose drawn with one continuous line. That ended up being the piece I was most satisfied with that week.

Moving forward, I appreciated that the course also addressed drawing fundamentals outside of VR. Practicing in class motivated me to explore more on my own, studying how to draw figures, understanding proportions, and learning how the human body works structurally. This was the part of the semester I found myself genuinely invested in. There was something satisfying about figuring out how to construct a figure step by step, rotating it mentally, understanding how it looks from different angles before translating it into 3D space. (Figure 2)

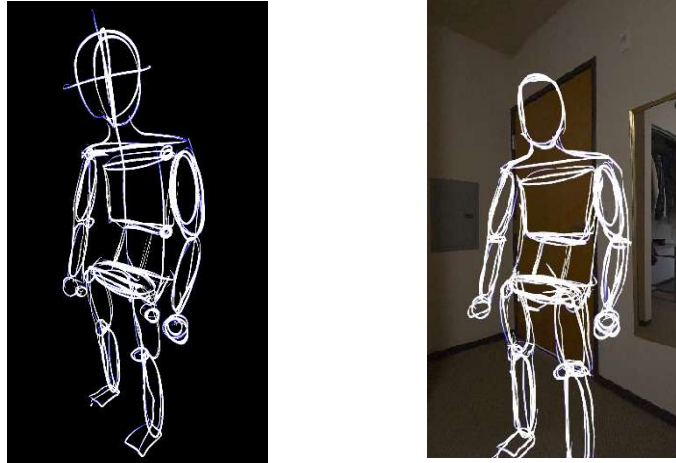


Figure 2. Human figure sketch

One of my favorite pieces came from a week focused on the expressive power of hands in drawing. This immediately made me think of the Creation of Adam. Before jumping into VR, I researched tutorials and practiced drawing hands on paper first. That preparation made the VR portion much more manageable. I already knew where to start and how to develop the drawing, and I think the result was one of the stronger pieces I produced all semester. (Figure 3)

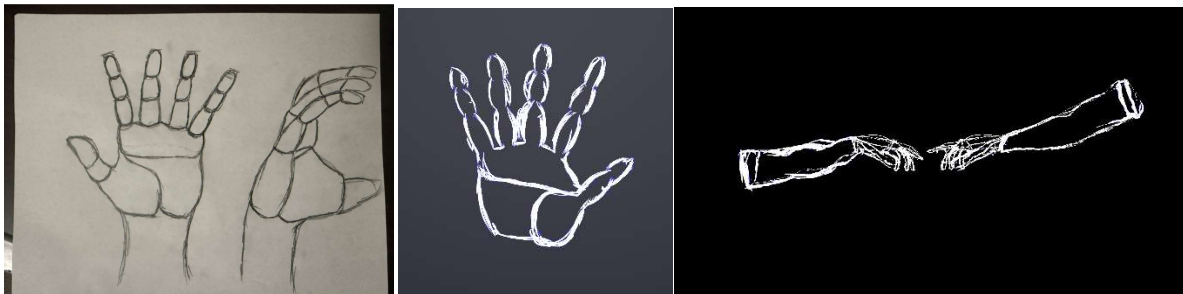


Figure 3. Drawing hands from paper to VR

The week where I drew a guitar using drawing from life was another that stood out. I was surprised by how closely my drawing resembled the actual instrument when using it as a live reference. Drawing from a real object in real time changed how I approached the whole piece. It felt more intentional and grounded than working from memory. (Figure 4)

The final weekly drawing before starting the final project was an interesting experiment. I was referencing *Stranger Things*, trying to build out an

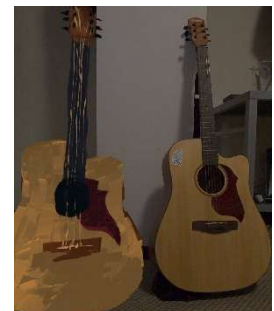


Figure 4. Guitar replicate with drawing from life

environment by importing 3D files and combining them with hand-drawn elements. It did not fully come together the way I imagined, but the process of trying to integrate different elements in one scene taught me a lot about composition and layering in VR.

Conclusion

Looking back, the biggest shift was not really in technical skill alone. It was in how I approached drawing as a practice. Early in the semester I would just start and see what happened. By the end I was researching, practicing on paper, studying references, and planning before picking up a controller. VR drawing pushed me to take the fundamentals of drawing more seriously than I expected going in.

2. Final Project Documentation

What the project is

My final project (Living Strokes) is a mixed reality drawing tool that I built for the Meta Quest 3. The core idea is that a drawing does not have to be static. You draw a stroke in the air with the right controller, and then you are given a short window to perform a physical gesture with your arm. That gesture gets recorded and embedded into the stroke. From that point on, the stroke animates continuously based on the movement you gave it, looping forever. Every stroke in the scene holds a different gesture from a different moment, all moving at the same time.

The experience is interactive by design. Users are not just looking at something someone else made, they are making something themselves, giving it life through their own body movement, and watching it persist in the space in front of them.

Creative process

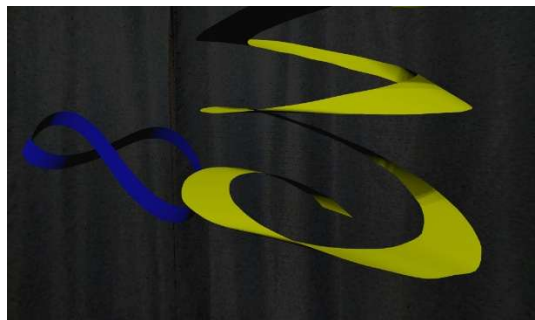


Figure 5. First attempt on animating the drawings by having a looped movement

I actually started the project earlier than the final project period. For a weekly drawing assignment, I built a basic prototype using the CavePainting codebase where strokes would slowly shift colors and move up and down after being drawn. It was simple, but it planted the

core idea for the final result: what if a drawing did not have to stay still after you made it? (Figure 5)

That experiment became the starting point for Living Strokes. The question became more specific from there: instead of generic movement, what if the animation came from the person who drew it? What if their body movement shaped how the stroke lived?

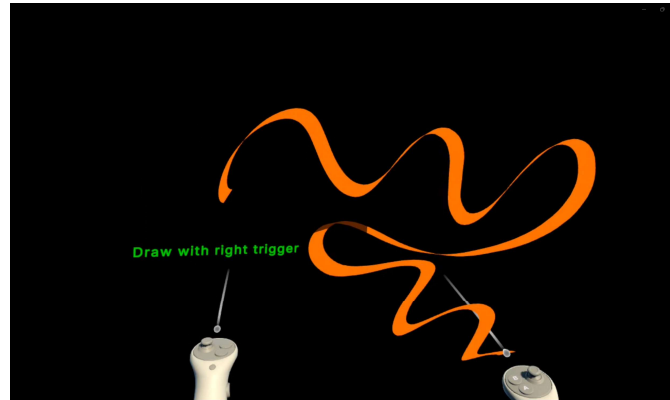


Figure 6. Making a drawing

The technical foundation came from the CavePainting Lite codebase provided in the course, which gave me a working VR ribbon stroke system to build on top of. In Figure 6, notice the orange ribbon strokes floating in the air above the controller. Each stroke was drawn with the right controller trigger and remains in place after being made. The instruction text visible on the left side of the screen is part of the UI system that guides users through each step of the interaction pipeline. From there I built a gesture recording pipeline: after finishing a stroke, a countdown begins, a colored trail appears showing the hand's path in real time, and those positions get processed and baked into a motion texture. A custom vertex shader reads that texture on the GPU every frame and displaces the stroke's vertices to produce smooth, continuous animation without performance cost. In Figure 7, notice the cyan trail following the controller, which makes the gesture recording window visible and spatial in real time.

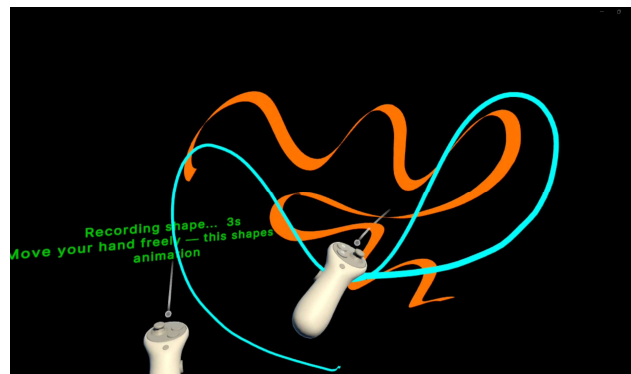


Figure 7. Gesture recording in progress

The interaction design went through many iterations. Early versions committed the gesture immediately with no way to preview or redo. That felt wrong for a tool that is meant to be expressive. The final version includes confirmation steps at every stage. You see your stroke before animating it, you preview the animation before committing to it, and you can redo any step independently. There is also an optional second gesture layer called rhythm, which controls how fast the animation cycles. A fast energetic movement produces a quick cycle and a slow deliberate movement produces something that breathes slowly.

Users can also choose between two animation modes before recording their gesture. Wave mode sends the gesture traveling from one end of the stroke to the other, like a ripple, but the pulse mode applies it to the whole stroke simultaneously. These two modes produce visually distinct results from the same gesture.

Most challenging aspect

The hardest problem was getting the gesture to actually feel present in the animation rather than just producing a generic wave. Early versions used raw position data directly, which caused strokes to drift across the room following wherever the hand had been. Later versions extracted frame-to-frame movement differences, which produced a jittery vibration rather than anything recognizable. The final approach normalizes the gesture by subtracting its straight-line drift and dividing by its maximum deviation, leaving only the pure character of the movement. That gets baked into a texture the shader samples over time. Getting this math right took significant iteration, and small parameter changes produced very different visual results.

Most successful aspect

The confirmation and preview system. What started as a purely technical project became genuinely expressive once users could see what they were making before committing to it. The preview step, where the stroke animates for a few seconds while the user decides whether to keep or redo the gesture, creates space for intention. It changes the relationship between the user and the tool. That interaction loop felt right in a way that the earlier automatic commitment did not.

Opportunities for future refinement

The most immediate opportunity is turning this into a more rigorous research project. The core concept, that a performer's gesture can be embedded into a drawing and expressed through it indefinitely, has not been explored in VR drawing tools in this specific way. Formalizing a research question around expressiveness and gesture-driven animation, running user studies, and comparing this approach to existing tools would be a natural next step.

On the technical side, the gesture-to-animation mapping could be more direct. Currently the system derives statistical properties from the gesture rather than replaying it precisely. A more literal translation of the recorded path would make the connection between body movement and

visual result more immediate. The visual quality of the strokes themselves also has room for improvement, as does color control, which was planned but not implemented in the final version.

A more ambitious future direction would be multi-stroke interaction, where strokes respond to each other's proximity or share gesture data to create coordinated movement across a whole scene. That would shift the experience from individual stroke animation toward something closer to a living ecosystem, which was my original conceptual ambition.

Observations from the final exhibit

Watching other presentations before my own was instructive. The projects that generated the strongest audience reactions were the ones where visitors could actually do something. Drawings and installations were interesting to look at, but the energy in the room was different when someone could put on a headset and interact with what was there. People leaned in differently. They wanted to try it themselves.

On my own project, the main thing I observed was that the interaction pipeline was not immediately intuitive for new users. The sequence of steps, draw, confirm, choose a mode, record a gesture, preview, optionally add rhythm, then finish, made complete sense to me after months of working on it. But for someone picking it up for the first time, there was a moment of confusion at almost every transition. People were not always sure what the text on their hand was asking them to do, or which button to press, or what the gesture was supposed to accomplish. The haptic feedback helped, but the conceptual gap between "record a gesture" and "watch it appear in your drawing" was harder to bridge in a live setting than I expected. That gap is probably the most important thing to address in any future version of the project.

3. Reflections on Observational 3D Drawing

Before this course I would not have said observation was a meaningful part of drawing. I thought drawing was about output and how it ended up on the page or screen.

The clearest example for me was when I used drawing from life to draw the guitar using the new feature. What surprised me was not that the drawing resembled the guitar at the end, it was how differently I looked at it once I was trying to draw it. I noticed the curve of the body, the proportions of the neck, the way the tuning pegs were arranged. I had owned a guitar for almost a year now and had never actually looked at it that carefully.

The same thing happened with when I was trying to draw the human figure. I spent some time studying how the human body is actually constructed, where joints are, how weight distributes, what the torso looks like from different angles. That study happened mostly on paper before I ever opened VR. But it carried directly into the 3D work and knowing what I was looking at made it possible to make decisions about what to draw and how to develop it.

Sighting was something I found useful but difficult to apply consistently. In VR, your reference point is your own body, which moves and that makes consistent measurement harder. The space itself shifts as you reposition, and what looked proportional from one angle can look wrong from another. I found myself relying more on intuition and visual checking from multiple viewpoints than on systematic sighting.

The question of whether software tools should explicitly support observation is one I find genuinely interesting after this semester. My instinct is yes, but carefully. Tools that impose observational structure can be valuable, like the reference import features and guide systems we explored in the course. Being able to bring a photograph into the virtual space and draw alongside it changed the quality of the output in a measurable way. The guitar drawing and the hands drawing both benefited directly from having a reference present.

What I would find most useful in a future tool is support for the transition between observation and execution. The moment where you look carefully at something and then have to hold that observation in mind while you draw it is where most errors happen. Anything that helps bridge that gap, whether through better reference positioning, annotation tools, or even just the ability to quickly toggle between reference and drawing view, would be genuinely useful without replacing the observational work itself.

After fifteen weeks, I think observation is not just a technique. It is a disposition. It is about choosing to look more carefully before you make a mark. VR drawing, at its best, created conditions where that choice felt necessary and rewarding.